SOFTWARE DESIGN SPECIFICATION (SDS)

Digital Barangay Blotter and Incidents Tracking

Project Title	**Digital Barangay Blotter and Incidents Tracking**
System Name	Incident Log
Document Version	v1.0
Date Prepared	03 May 2026
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Course	Software Verification and Validation
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DOCUMENT CONTROL

Version	**Date**	**Author**	**Description**
v1.0	03 May 2026	GIT HIVE - BSCS 3B	Initial release of SDS document

SRS Document	**Digital Barangay Blotter and Incidents Tracking Requirements**
SRS Version	v1.0
SRS Date	25 April 2026
Verification Method	Review, walkthrough, and client confirmation

1. Introduction

1.1 Purpose

This Software Design Specification (SDS) describes the architectural and detailed design of the Bari E Blotter system. The document serves as the technical blueprint for the implementation of the Digital Barangay Blotter and Incidents Tracking System for Barangay Bari. The SDS is intended for developers, project advisers, testers, and stakeholders to ensure that the system design satisfies the approved functional and non-functional requirements.

1.2 Scope

The system is a web-based application designed to manage barangay blotter records, patrol assignments, incident verification, and crime analytics. The design includes modules for authentication, incident management, reporting, dashboard analytics, and data export. The current version focuses on internal barangay operations and does not yet include citizen mobile reporting features.

1.3 Definitions, Acronyms, and Abbreviations

Term	Definition
SDS	Software Design Specification
SRS	Software Requirements Specification
UI	User Interface
DB	Database
RBAC	Role-Based Access Control
API	Application Programming Interface

1.4 References

• Digital Barangay Blotter and Incidents Tracking Requirements Document v1.0

• IEEE 1016-2009 Software Design Description Standard

• SWEBOK v3.0 Software Engineering Body of Knowledge

1.5 Design Overview

The Bari E Blotter system follows a three-layer client-server architecture composed of the presentation layer, application logic layer, and database layer. The presentation layer provides responsive web-based interfaces for the Barangay Captain and Auditor. The business logic layer processes incident validation, patrol assignments, report generation, and analytics. The database layer stores user accounts, blotter records, incident reports, and historical analytics data.

2. System Architecture

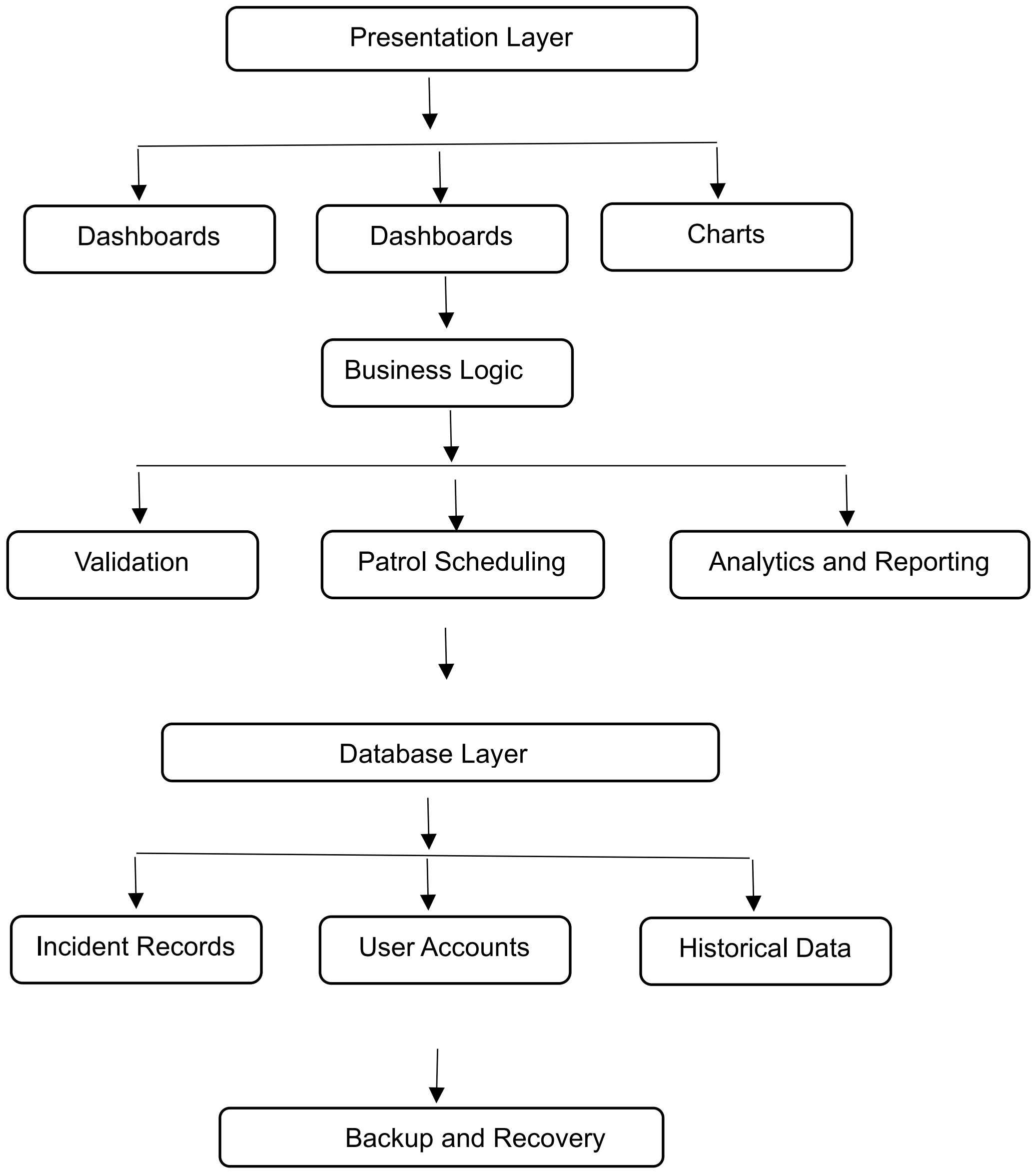
2.1 Architectural Pattern

Selected Pattern	Three-layer Client-Server Architecture with MVC Design

Justification	Supports modularity, security, maintainability, and scalability for barangay operation
Traceability	ADR-001 to ADR-006

2.2 System Architecture Diagram

The architecture consists of three major layers:



2.3 Component Description

Component Name	Responsibility	Technology
Authentication Module	Handles login, sessions, and RBAC	Node.js + Express.js
Incident Management Module	Records and validates incidents	Node.js + PostgreSQL
Patrol Scheduling Module	Assigns patrol teams to puroks	Node.js + Express.js

Analytics Dashboard	Displays charts and summaries	Chart.js
Database Layer	Stores all application records	PostgreSQL

2.4 Technology Stack

Layer	Technology	Version	Justification
Frontend/UI	Javascript	Latest	Responsive web interface
Backend	Node.js + Express.js	10.x	Secure and modular framework
Database	PostgreSQL	8.x	Reliable relational database
Authentication	JWT + bcrypt	N/A	Role separation and security
Hosting	Vercel (Frontend) + Render (Backend & Database)	N/A	Accessible during development
Version Control	Git/GitHub	Latest	Team collaboration and backups

3. Detailed Design

3.1.1 Authentication Module

Field	Description
Component Name	Authentication Module
SRS Traceability	FDR-001 to FDR-004
Responsibility	Handles secure login and access control.
Input/Preconditions	Authenticated user access and valid request data
Output/Postconditions	Updated database records and generated responses
Processing Logic	Validates data, executes business rules, and stores results

Error Handling	Displays validation messages and logs exceptions

3.1.2 Incident Management Module

Field	Description
Component Name	Incident Management Module
SRS Traceability	FDR-006 to FDR-012
Responsibility	Manages recording and verification of incidents.
Input/Preconditions	Authenticated user access and valid request data
Output/Postconditions	Updated database records and generated responses
Processing Logic	Validates data, executes business rules, and stores results
Error Handling	Displays validation messages and logs exceptions

3.1.3 Patrol Scheduling Module

Field	Description
Component Name	Patrol Scheduling Module
SRS Traceability	FDR-005
Responsibility	Handles patrol assignment scheduling.
Input/Preconditions	Authenticated user access and valid request data
Output/Postconditions	Updated database records and generated responses
Processing Logic	Validates data, executes business rules, and stores results
Error Handling	Displays validation messages and logs exceptions

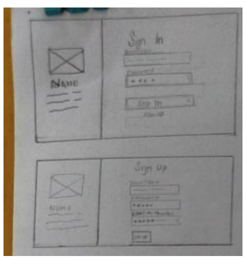
3.1.4 Analytics and Reporting Module

Field	Description
Component Name	Analytics and Reporting Module
SRS Traceability	FDR-013 to FDR-017

Responsibility	Generates charts, reports, and exports.
Input/Preconditions	Authenticated user access and valid request data
Output/Postconditions	Updated database records and generated responses
Processing Logic	Validates data, executes business rules, and stores results
Error Handling	Displays validation messages and logs exceptions

3.2 User Interface Design

3.2.1



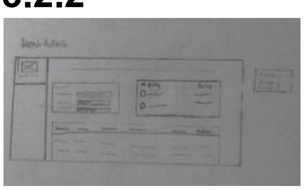
Sign In Screen: Sign in Screen

Triggered by: User launching the application or clicking the "Sign In" link from the Sign Up page

SRS traceability: REQ-004 (User Authentication), FR-01 (System Access Control)**.**

Description: split-screen design so the branding stays on the left side while the user switches between signing in and signing up on the right. This keeps the look consistent and makes the app easier to use.

3.2.2



Sign In Screen: Admin Dashboard

Triggered by: Successful user login from the Sign In screen.

SRS traceability: FR-09 (Incident Visualization), FR-11 (Incident Management).

Description: The dashboard uses a Sidebar Navigation layout. The main content area is divided into a top section with visual summaries charts and a bottom section for the detailed incident list.

3.2.3

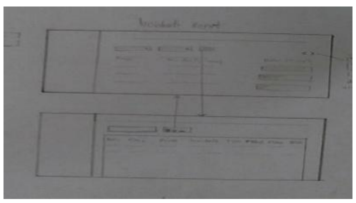
Sign In Screen: Admin Verification Screen

Triggered by: Admin clicking the "Verification" or "Pending Tasks" link in the Sidebar.

SRS traceability: FR-12 (Report Verification), FR-11 (Incident Management)

Description: he screen maintains the Sidebar Navigation on the left for consistency. The main area features a large Verification Table where incoming reports are queued for review.

3.2.4



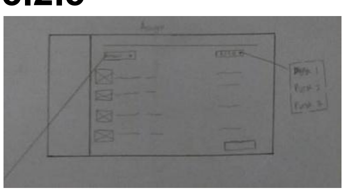
Sign In Screen: Incident Report Screen

Triggered by: Admin selecting "Incident Report" from the sidebar or clicking a specific record for more details.

SRS traceability: Admin selecting "Incident Report" from the sidebar or clicking a specific record for more details.

Description: This interface uses a Dual-View Transition. The top box represents the Detailed View (form/summary style), and the bottom box represents the List View (tabular style). Arrows indicate that selecting a row in the table opens the detailed summary above.

3.2.5



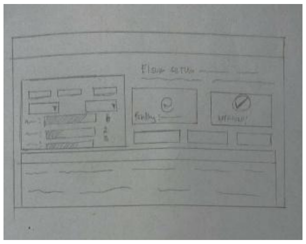
Sign In Screen: Assign Area Screen

Triggered by: Admin selecting "Assign" or "Personnel Management" from the Sidebar.

SRS traceability: FR-14 (User Role Assignment), FR-15 (Purok Management)

Description: This screen features a List-to-Dropdown interaction. On the left side of the main panel, there is a list of registered personnel with their photos/avatars. On the right, there are dropdown menus used to select which Purok they will manage.

3.2.6



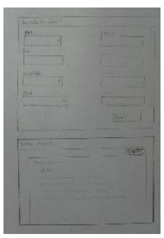
Sign In Screen: Audit Dashboard Screen

Triggered by: Admin or Auditor selecting the "Audit" or "Summary" view from the navigation.

SRS traceability: FR-09 (Visualization), FR-16 (System Audit/Statistics)

Description: This screen uses a Dashboard Grid layout. It features a mini version of the incident bar chart on the left and large Status Cards in the center-right to show overall system progress.

3.2.7



Sign In Screen: Incident Report

Triggered by: User clicking "Create New Report" or "Blotter Entry" from the dashboard.

SRS traceability: FR-09 (Visualization), FR-16 (System Audit/Statistics)

Description: The design consists of two main layouts:

1. **Top Box:** A structured form with specific dropdowns and text fields for quick reporting.
2. **Bottom Box:** A "Blotter Record" view, designed like a formal document or digital notebook for detailed narratives.

3.3 Interface Design

Caller Component	Called Component	Data Passed	Expected Return
Dashboard	Incident Module	Incident Request	Incident Records
Incident Module	Database	SQL Queries	Stored Data

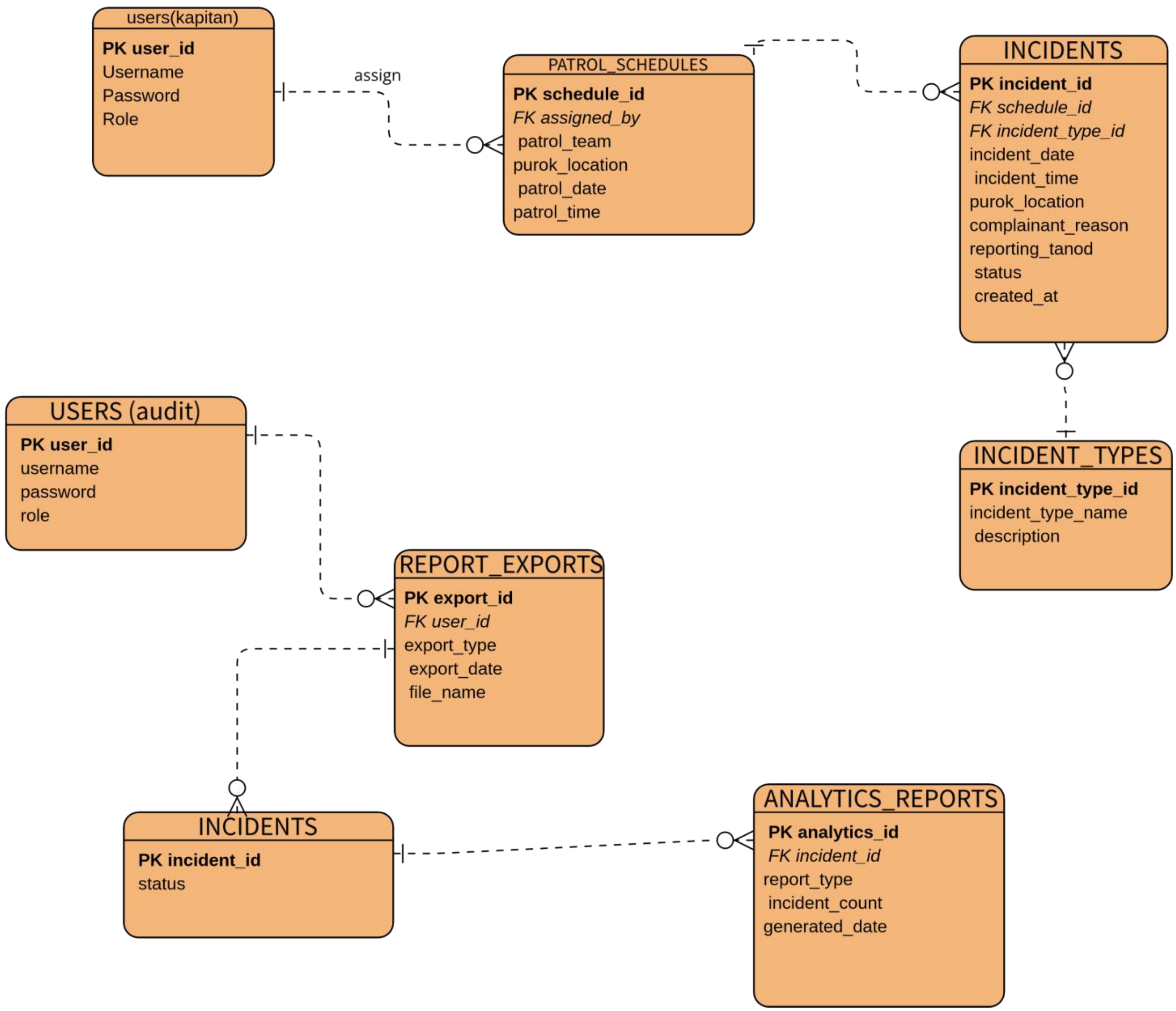
Analytics Module	Database	Incident Statistics	Chart Data

External System	Purpose	Interface Type	Reference
MySQL Database	Data storage	SQL	DDR-001
Chart.js	Visualization	JavaScript Library	FDR-013

4. Data Design

4.1 Entity Relationship Overview

The database design includes entities for Users, Incident Records, Patrol Schedules, Login Logs, Incident Types, and Reports. Relationships are designed to maintain referential integrity and support historical archiving.



4.2 Data Dictionary

Entity/Table: Users

Attribute	Data Type	Constraints	Key	Description
Attribute Name	**Data Type**	**Constraints**	**Key**	**Description**
user_id	INT	NOT NULL, AUTO_INCREMENT	PK	Unique identifier for each user
username	VARCHAR(100)	NOT NULL, UNIQUE		Username used during login
password	VARCHAR(255)	NOT NULL		Encrypted password using secure hashing
role	VARCHAR(50)	NOT NULL		User role (Captain or Auditor)
created_at	DATETIME	NOT NULL		Timestamp of account creation

Entity / Table: LOGIN_LOGS

Attribute Name	Data Type	Constraints	Key	Description
log_id	INT	NOT NULL, AUTO_INCREMENT	PK	Unique login log identifier
user_id	INT	NOT NULL	FK	References USERS(user_id)
login_time	DATETIME	NOT NULL		Date and time of login attempt
login_status	VARCHAR(20)	NOT NULL		Login result (Success or Failed)

Entity / Table: PATROL_SCHEDULES

Attribute Name	Data Type	Constraints	Key	Description
schedule_id	INT	NOT NULL, AUTO_INCREMENT	PK	Unique patrol schedule identifier
assigned_by	INT	NOT NULL	FK	References USERS(user_id)
patrol_team	VARCHAR(100)	NOT NULL		Assigned patrol team
purok_location	VARCHAR(100)	NOT NULL		Patrol area location
patrol_date	DATE	NOT NULL		Scheduled patrol date
patrol_time	TIME	NOT NULL		Scheduled patrol time
created_at	DATETIME	NOT NULL		Timestamp of schedule creation

Entity/Table: Incidents

Attribute	Data Type	Constraints	Key	Description
incident_id	INT	PK, AUTO_INCREMENT	Primary Key	Unique incident identifier
incident_type	VARCHAR(100)	NOT NULL		Incident category
location	VARCHAR(100)	NOT NULL		Purok location
status	VARCHAR(30)	NOT NULL		Pending, Verified, Rejected

Entity / Table: INCIDENTS

Attribute Name	Data Type	Constraints	Key	Description
incident_id	INT	NOT NULL, AUTO_INCREMENT	PK	Unique incident record identifier
incident_type_id	INT	NOT NULL	FK	References INCIDENT_TYPES(incident_type_id)
schedule_id	INT	NOT NULL	FK	References PATROL_SCHEDULES(schedule_id)
incident_date	DATE	NOT NULL		Date when incident occurred
incident_time	TIME	NOT NULL		Time when incident occurred
purok_location	VARCHAR(100)	NOT NULL		Location of the incident
complainant_reason	TEXT	NOT NULL		Complaint details and incident reason
reporting_tanod	VARCHAR(100)	NOT NULL		Assigned reporting tanod
status	VARCHAR(30)	NOT NULL		Incident status (Pending, Verified, Rejected)
created_at	DATETIME	NOT NULL		Timestamp of incident creation

Entity / Table: REPORT_EXPORTS

Attribute Name	Data Type	Constraints	Key	Description
export_id	INT	NOT NULL, AUTO_INCREMENT	PK	Unique export identifier
user_id	INT	NOT NULL	FK	References USERS(user_id)
export_type	VARCHAR(20)	NOT NULL		Export file format (CSV or PDF)
export_date	DATETIME	NOT NULL		Date and time of export
file_name	VARCHAR(255)	NOT NULL		Name of exported report file

Entity / Table: ANALYTICS_REPORTS

Attribute Name	Data Type	Constraints	Key	Description
analytics_id	INT	NOT NULL, AUTO_INCREMENT	PK	Unique analytics report identifier
incident_id	INT	NOT NULL	FK	References INCIDENTS(incident_id)

report_type	VARCHAR(50)	NOT NULL		Analytics category (Daily, Weekly, Monthly, Yearly)
incident_count	INT	NOT NULL		Total incidents counted
generated_date	DATETIME	NOT NULL		Timestamp of analytics generation

5. Security Design

5.1 Authentication and Authorization

Authentication Mechanism	[e.g., Username + password with bcrypt hashing / JWT tokens / OAuth 2.0 / session-based]

SRS Traceability	[Requirement ID(s) from the verified SRS]

Authorization Model	[e.g., Role-based access control (RBAC) / Attribute-based / no multi-role]

Roles Defined	[List all user roles and their permissions — e.g., Admin: full access; User: read own data only]

Security Design Detail
[Describe how authentication and authorization are implemented. Where are tokens stored? How are passwords hashed? How are roles enforced — at the controller level, middleware, or database?]

5.2 Input Validation and Injection Prevention

Validation Location	[Server-side, client-side, or both?]

Validation Approach	[e.g., Allowlist validation, parameterized queries for all DB access, ORM]

Threats Addressed	[e.g., SQL injection via parameterized queries; XSS via output encoding; CSRF via CSRF tokens]

Validation Rules
[List specific validation rules for key inputs — e.g., Email: RFC 5322 format, max 255 chars; Password: min 8 chars, at least one uppercase, digit, and special character; Age: integer 0–120]

5.3 Data Protection

Data Encrypted in Transit	[Protocol — e.g., HTTPS/TLS 1.3 enforced for all communications]

Data Encrypted at Rest	[e.g., Sensitive fields encrypted using AES-256; or 'not applicable for this project scope']

Sensitive Data Identified	[List data elements that are sensitive: passwords, payment info, personal data, health data]

Logging Policy	[What is logged? What is explicitly NOT logged (passwords, tokens, PII)?]

6. Requirements Traceability Matrix

Req. ID	Requirement Summary	Design Element(s)	Section Ref.	Status
FR-1	Provide separate roles for Admin (Captain) and Auditor.	Role-Based Access Control (RBAC) Module	§ 5.1	**Addressed**
FR-2	Validate user credentials (Username & Password).	Sign In Screen / Login Validator	§ 3.2.1	**Addressed**
FR-3	Record User ID and time of every login attempt.	Login Audit Log Table	§ 5.3	**Addressed**

FR-4	Allow Captain to assign patrol teams to specific Puroks.	Assign Area Screen / Personnel List	§ 3.2.6	**Addressed**
FR-4.1	Admin view-only mode for incident reports to ensure transparency.	Read-Only Dashboard Table	§ 3.2.3	**Addressed**
FR-4.2	Allow Captain to verify or reject reported incidents.	Verification Screen / Action Buttons	§ 3.2.4	**Addressed**
FR-4.3	Support incident statuses: Pending, Verified, and Rejected.	Incident Table / Status Column	§ 3.2.4	**Addressed**
FR-5	Auto-tag patrol teams when an incident is recorded in their area. .	Automated Patrol Tagging Logic	§ 3.1.2	**Addressed**
FR-6	Show time, date, and purok location of recorded incidents.	Incident Report Screen	§ 3.2.5	**Addressed**
FR-6.1	Generate data tables filtered by day, week, month, and year.	Dashboard Search & Filter Bars	§ 3.2.3	**Addressed**
FR-7	Dropdown selection for categorizing incident types.	Incident Report Form / Dropdown	§ 3.2.8	**Addressed**
FR-9	Bar chart turns Red for >= 5 incidents and resets weekly.	get_chart_color() Function / Reset Job	§ 3.1.1	**Addressed**
FR-9.1	Allow filtering of areas to help Captain enhance security.	Audit Dashboard / Filter Controls	§ 3.2.7	**Addressed**
FR-10	Record detailed blotter reports and complainant reasons.	Digital Blotter Record / Narrative Area	§ 3.2.8	**Addressed**
FR-11	Ability to manually export database as CSV or PDF files.	Data Export Module	§ 4.2	**Addressed**
FR-12	Allow developer to restore database to a specific date/time.	Database Backup/Restore Utility	§ 5.3	**Addressed**

RTM SUMMARY

Total verified requirements in SRS	**12**
Requirements addressed in this SDS	**12**
Requirements NOT yet addressed (gaps)	**0**
Requirements NOT yet addressed (gaps)	**100%**

7. Design Assumptions and Constraints

7.1 Assumptions

Assumption	Impact if Incorrect
The Barangay Captain, Auditor and assigned personnel have basic knowledge of using web-based systems and dashboards.	The user interface and navigation flow may need to be redesigned to become simpler and more user-friendly, including additional training modules or guided navigation.
Incident reports will be encoded correctly and completely by authorized barangay personnel.	Additional validation controls, automated checking, and stricter form requirements may need to be redesigned to prevent inaccurate analytics and reports.
Patrol schedules are properly assigned before incidents occur within the purok areas.	The patrol assignment and incident auto-tagging logic may need redesign to support dynamic reassignment and manual patrol mapping
Stable internet or local network connectivity is available during normal operations.	Offline support, local caching, and synchronization mechanisms may need to be redesigned and implemented
Modern computers and updated web browsers are available in the barangay office.	The frontend system may need optimization or redesign for lower hardware specifications and older browser compatibility.
Authorized users will follow proper password confidentiality and account security practices.	Additional security controls such as multi-factor authentication, account monitoring, and stricter password policies may need to be redesigned.
Backup and export operations will be regularly performed by the assigned Auditor or system administrator.	Automated backup scheduling and recovery management features may need to be redesigned and integrated.
Incident categories and blotter classifications will remain manageable using predefined dropdown options.	The incident categorization module may need redesign to support customizable and expandable categories.
The expected number of incident records will remain manageable for the selected database system.	Database optimization, indexing, and scalable architecture redesign may become necessary

Only authorized barangay personnel will have access to system devices and credentials.	Additional device restrictions, audit logging, and security monitoring systems may need redesign and implementation.

7.1 Constraints

Type	Constraint	Design Response
Technical	The system must support patrol scheduling and purok-based monitoring.	The design includes patrol scheduling modules, location-based assignment, and incident auto-tagging functionality
Technical	The system must securely store and process incident records and confidential barangay information.	The design implements encrypted passwords, secure login sessions, and protected database access mechanisms.
Technical	The system must support role-based access control for Barangay Captain and Auditor users.	The design includes authentication modules, session management, and role-based permission validation.
Technical	The system must generate reports, analytics, and visual dashboards.	The design integrates filtering tools, analytics modules, chart visualization, and report export functions.
Security	Protect sensitive incident data	RBAC and encrypted passwords used
Security	The system must protect sensitive barangay records, login credentials, and personal complainant information from unauthorized access, modification, or data breaches.	The design implements encrypted passwords, secure session management, role-based access control, audit logging, authentication validation, and restricted database permissions to strengthen system security and data confidentiality.
Resource	The project has limited development time because of academic deadlines and research schedules.	The design prioritizes essential blotter management, analytics, and scheduling functionalities using modular development.
Resource	The project has limited financial and infrastructure resources.	The design uses open-source technologies, lightweight frameworks, and cost-effective deployment solutions.
Environmental	The system will operate within the barangay office environment using available local computers and network infrastructure.	The design follows browser-based implementation and lightweight web architecture for compatibility with existing resources.
Environmental	Possible interruptions such as unstable electricity or network connectivity may occur during operation.	he design includes backup and recovery support, auto-save concepts, and export functionalities to reduce data loss risks.

8. Appendices

Appendix A — Architecture Diagram Source File

Appendix B — UML Diagrams

Appendix C — Design Review Checklist

#	Checklist Item	Status
1	Every verified requirement from the SRS appears in the RTM (FR-1 through FR-13).	[☐] / [☑]
2	RTM coverage percentage is 100% (All functional and logic requirements are mapped).	[☐] / [☑]
3	Architecture diagram is present, labeled, and has a caption.	[☐] / [☑]
4	Every component in the architecture diagram (UI, Server, DB) has a description.	[☐] / [☑]
5	Every screen (Sign In, Sign Up, Dashboard, Verification, Incident Report, Assign, Audit, Blotter) has a wireframe and description.	[☐] / [☑]
6	ERD is present, labeled, and all entities (Puroks, Incidents, Weekly_History) have data dictionary entries.	[☐] / [☑]
7	All security requirements (Authentication, RBAC, Validation, Data Protection) are addressed in Section 5.	[☐] / [☑]
8	All placeholder text [in brackets] has been replaced with actual content (e.g., specific rules for Red color logic).	[☐] / [☑]
9	Revision history is up to date (Version 1.0, dated today).	[☐] / [☑]
10	Document has been spell-checked and reviewed by all team members (Git Hive - BSCS 3B).	[☐] / [☑]

Appendix D — Glossary of Design Patterns

For reference, the following design patterns are commonly applied in student projects at this level:

Pattern	Description	When to Use

MVC (Model-View-Controller)	Separates data (Model), display (View), and logic (Controller) into three distinct layers.	Used as the main architecture. The **Model** handles the incident database, the **View** shows the bar chart, and the **Controller** runs the "Red" color logic.
Observer	Defines a one-to-many dependency so that when one object changes state, all dependents are notified automatically.	Used for the **Live Alert System**. When an incident is approved, the Bar Chart is "notified" to update its height and check if it needs to turn Red.
Singleton	Ensures a class has only one instance and provides a global access point to it.	Used for the **Database Connection**. It ensures the system doesn't open too many connections to the incident database at once.
Facade	Provides a simplified interface to a complex subsystem.	Used for the **Report Generator**. It hides the complex PDF/CSV formatting logic behind a simple "Export" button for the Auditor.
Repository Pattern	Abstracts data access behind an interface so the app doesn't depend on one specific DB technology.	Used to separate the Incident logic from the SQL queries, making it easier to upgrade the database in the future.

References

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